# Critical inradius asymptotics and a supremum-norm bound for convex domains

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**Abstract**

We determine the leading large-exponent asymptotic of the critical inradius-normalized Dirichlet frequency constant. We also give an explicit lower bound for the supremum-norm problem in a separate range of exponents. These results address parts of AIM 1016 and do not determine its general sharp constants.

**Repository target.** This manuscript concerns AIM problem 1016 at revision `3fa50884d129`. Its scope is stated below.

## Result and scope

For a bounded nonempty open convex domain $\Omega \subset \mathbb{R}^N$, let $R_\Omega$ be its Euclidean inradius. In the critical case considered here, the constant in AIM 1016 is

$$C_{\mathrm{opt}}(N,N,q) = \inf_{\Omega \text{ convex}} R_\Omega^{N^2/q} \inf_{\substack{u \in C_c^\infty(\Omega) \\ u \neq 0}} \frac{\int_\Omega |\nabla u|^N \, dx}{\left(\int_\Omega |u|^q \, dx\right)^{N/q}}.$$

Equivalently, impose $R_\Omega = 1$ and take the infimum of the displayed quotient. Write

$$\sigma_{N-1} = |S^{N-1}|, \qquad N' = \frac{N}{N-1}.$$

For every integer $N \geq 2$, the following asymptotic holds:

$$\boxed{\lim_{q\to\infty} q^{N-1} C_{\mathrm{opt}}(N,N,q) = \sigma_{N-1} \left(\frac{eN^2}{N-1}\right)^{N-1}.} \tag{1}$$

In particular,

$$\lim_{q\to\infty} q\, C_{\mathrm{opt}}(2,2,q) = 8\pi e.$$

This determines an asymptotic regime of the parameter-dependent problem, **not** the exact constant at a fixed finite $q$. The proof is a consequence of classical sharp exponential integrability, a uniform convex-domain Hardy estimate, and an explicit concentrating test. No claim that the asymptotic is new in the literature is made.

## 1. A shape-uniform L^N estimate

By scaling, restrict to domains of inradius one. Their boundary-distance function $d(x) = \operatorname{dist}(x, \partial\Omega)$ satisfies $0 < d \le 1$, $|\nabla d| = 1$ almost everywhere, and is concave because the domain is convex. In particular $\Delta d \le 0$ distributionally.

For $u \in C_c^\infty(\Omega)$, integration by parts using

$$\operatorname{div}\left(\frac{\nabla d}{d^{N-1}}\right) \le -\frac{N-1}{d^N}$$

and then Hölder's inequality give the standard convex Hardy estimate

$$\left\|\frac{u}{d}\right\|_N \le \frac{N}{N-1}\|\nabla u\|_N.$$

The distributional computation can equivalently use the locally BV field $\nabla d/(d+\varepsilon)^{N-1}$, whose singular divergence part is nonpositive, and then let $\varepsilon \downarrow 0$. Adding $\varepsilon$ regularizes the denominator, not the distance function itself. Since $d \le 1$, this implies

$$\|u\|_N^N \le B_N \|\nabla u\|_N^N, \qquad B_N = \left(\frac{N}{N-1}\right)^N. \tag{2}$$

The constant is uniform over the entire inradius-one class; no bound on volume or diameter is needed.

## 2. Exponential integrability and moment bounds

Use the Adachi–Tanaka inequality on $\mathbb{R}^N$. Put

$$\alpha_N = N\sigma_{N-1}^{1/(N-1)}, \qquad \Phi_N(t) = e^t - \sum_{j=0}^{N-2} \frac{t^j}{j!}.$$

For each fixed $0 < \alpha < \alpha_N$, there is a finite $A_{N,\alpha}$ such that every $u \in W^{1,N}(\mathbb{R}^N)$ with $\|\nabla u\|_N = 1$ satisfies

$$\int_{\mathbb{R}^N} \Phi_N(\alpha|u|^{N'})\,dx \le A_{N,\alpha}\|u\|_N^N. \tag{3}$$

This is Theorem 0.1 of the authors' 1999 primary manuscript and the principal theorem of their 2000 journal paper:

- S. Adachi and K. Tanaka, *A scale-invariant form of Trudinger–Moser inequality and its best exponent*, RIMS Kokyuroku **1102** (1999), 148–153. Primary PDF: https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/pdf/1102-11.pdf
- S. Adachi and K. Tanaka, *Trudinger type inequalities in R^N and their best exponents*, Proc. Amer. Math. Soc. **128** (2000), 2051–2057. DOI: https://doi.org/10.1090/S0002-9939-99-05180-1

Extend a compactly supported domain test by zero to $\mathbb{R}^N$ and normalize its gradient norm to one. Combining (2) and (3), and retaining the $m$th nonnegative exponential-series term for any integer $m \ge N-1$, gives

$$\int_\Omega |u|^{N'm}\,dx \le A_{N,\alpha}B_N\, m!\,\alpha^{-m}. \tag{4}$$

Set $r_m = N'm$. Taking the power $N/r_m = (N-1)/m$ in (4), and then taking the reciprocal, proves the shape-uniform lower bound

$$C_{\mathrm{opt}}(N,N,r_m) \ge \alpha^{N-1}(A_{N,\alpha}B_N)^{-(N-1)/m}(m!)^{-(N-1)/m}. \tag{5}$$

Stirling's formula yields

$$\liminf_{m\to\infty} r_m^{N-1} C_{\mathrm{opt}}(N,N,r_m) \ge (eN'\alpha)^{N-1}. \tag{6}$$

This also holds along all real $q \to \infty$, not only the sequence $r_m$. Indeed, choose $m = \lceil q/N' \rceil$, so $q \le r_m < q + N'$, and interpolate between the uniform $L^N$ estimate (2) and (4):

$$\frac{1}{q} = \frac{\theta_q}{N} + \frac{1-\theta_q}{r_m}, \qquad \theta_q = \frac{N(r_m - q)}{q(r_m - N)} = O(q^{-2}).$$

Thus $\|u\|_q^N \le B_N^{\theta_q}\|u\|_{r_m}^{N(1-\theta_q)}$; the interpolation factors are asymptotically one after division by $q^{N-1}$. Taking the infimum over all shapes and tests preserves this uniform bound. Finally let $\alpha$ increase to $\alpha_N$ in (6), obtaining

$$\liminf_{q\to\infty} q^{N-1} C_{\mathrm{opt}}(N,N,q) \ge (eN'\alpha_N)^{N-1} = \sigma_{N-1}\left(\frac{eN^2}{N-1}\right)^{N-1}. \tag{7}$$

The order of limits is important: $\alpha < \alpha_N$ is fixed while $q$ tends to infinity. No uniform control of $A_{N,\alpha}$ at the critical exponent is assumed.

## 3. A matching Moser test in the unit ball

The unit ball has inradius one, so its test functions provide upper bounds for the shape infimum. For $L > 0$, define the radial function

$$u_L(r) = \sigma_{N-1}^{-1/N}\begin{cases} L^{(N-1)/N}, & 0 \le r \le e^{-L}, \\ L^{-1/N}\log(1/r), & e^{-L} < r < 1, \\ 0, & r \ge 1. \end{cases}$$

This belongs to $W_0^{1,N}(B_1)$, and direct integration gives

$$\int_{B_1} |\nabla u_L|^N\,dx = 1. \tag{8}$$

Its inner plateau alone yields

$$\int_{B_1} |u_L|^q\,dx \ge \frac{\sigma_{N-1}}{N} e^{-NL}\sigma_{N-1}^{-q/N} L^{q(N-1)/N}.$$

Therefore

$$C_{\mathrm{opt}}(N,N,q) \le \sigma_{N-1}\left(\frac{N}{\sigma_{N-1}}\right)^{N/q} e^{N^2L/q} L^{-(N-1)}. \tag{9}$$

The nonsmooth test is legitimate by smooth approximation in $W_0^{1,N}(B_1)$ and $L^q(B_1)$ for each finite $q$. Alternatively, place the same test inside a fixed smaller concentric ball and let its radius tend to one.

Choose

$$L = \frac{(N-1)q}{N^2}.$$

Then (9) gives

$$\limsup_{q\to\infty} q^{N-1} C_{\mathrm{opt}}(N,N,q) \le \sigma_{N-1} \left(\frac{eN^2}{N-1}\right)^{N-1}. \tag{10}$$

Equations (7) and (10) prove (1).

## Additional bound for the supremum norm

### Statement and scope

Let $N \ge 2$. For a nonempty bounded open convex set $\Omega \subset \mathbb{R}^N$, define its Euclidean inradius and its Dirichlet Poincaré–Sobolev frequency at $q = \infty$ by

$$R_\Omega = \sup\{r > 0 : \text{there is } x \in \mathbb{R}^N \text{ with } B_r(x) \subset \Omega\},$$

$$\lambda_{p,\infty}(\Omega) = \inf_{\substack{f \in C_c^\infty(\Omega;\mathbb{R}) \\ f \neq 0}} \frac{\displaystyle\int_\Omega |\nabla f|^p \, dx}{\|f\|_{L^\infty(\Omega)}^p}.$$

For $p > N$, the constant requested in this parameter regime of Problem 1016 is

$$C_{\mathrm{opt}}(N,p,\infty) = \inf_{\substack{\Omega \subset \mathbb{R}^N \\ \Omega \text{ nonempty, bounded, open, convex}}} R_\Omega^{p-N} \lambda_{p,\infty}(\Omega). \tag{S1}$$

Scaling gives the equivalent definition with the constraint $R_\Omega = 1$. The following explicit lower bound holds in the range $p \ge N+1$:

$$\boxed{C_{\mathrm{opt}}(N,p,\infty) \ge 2\left(\frac{p-N}{p-1}\right)^{p-1} \frac{\pi^{(N-1)/2}\Gamma((p-N+1)/2)}{\Gamma(p/2)}.} \tag{S2}$$

For example, (S2) gives the exact comparison

$$C_{\mathrm{opt}}(2,4,\infty) \ge \frac{8\pi}{27}.$$

This is a lower bound, not a determination of the optimal constant. No equality case, minimizing domain, or sharpness assertion is made. It adds a parameter regime different from the critical-exponent asymptotic $p = N$, $q \to \infty$. The proof below does not cover $N < p < N+1$, and no claim of bibliographic priority is made.

## 1. Balanced supporting planes

Normalize $R_\Omega = 1$, and translate a center of a maximal inscribed ball to the origin. Such a center exists because $\Omega$ is bounded. There are contact supporting planes with unit outward normals $n_1, \dots, n_m$, and positive weights $\lambda_1, \dots, \lambda_m$, such that

$$\Omega \subset P := \bigcap_{i=1}^{m} \{x : n_i \cdot x < 1\}, \qquad \sum_{i=1}^{m} \lambda_i = 1, \qquad \sum_{i=1}^{m} \lambda_i n_i = 0. \tag{S3}$$

Here one can take $m \le N+1$. To justify the balance condition, let $h_\Omega(n) = \sup_{x\in\Omega} n \cdot x$, and consider the compact set of contact normals satisfying $h_\Omega(n) = 1$. If their convex hull did not contain zero, strict separation would give a direction in which moving the ball center increases the distance from all contact planes. The support function is continuous, so the same strict gain holds in a neighborhood of the contact normals; away from that neighborhood the support distances already exceed one by a positive amount. A sufficiently small displacement would therefore permit a larger inscribed ball, a contradiction. Carathéodory's theorem gives the finite collection in (S3); zero weights can be discarded.

Fix a nonzero test function $f \in C_c^\infty(\Omega; \mathbb{R})$. Choose a point $x_* \in \Omega$ at which its absolute value attains its maximum, and put

$$M = |f(x_*)| = \|f\|_\infty, \qquad a_i = 1 - n_i \cdot x_* > 0, \qquad v_i = \frac{n_i}{a_i}, \qquad \rho_i = \lambda_i a_i.$$

The identities in (S3) imply

$$\sum_i \lambda_i a_i = 1, \qquad \sum_i \rho_i = 1, \qquad \sum_i \rho_i v_i = 0. \tag{S4}$$

Thus the $\rho_i$ are probability weights, and the vectors $v_i$ have weighted mean zero. In particular, the maximum point $x_*$ need not be an incenter.

## 2. A support-function estimate

Write

$$s = p - N \ge 1, \qquad h(\omega) = \max_i (v_i \cdot \omega), \quad \omega \in S^{N-1}.$$

Equation (S4) implies $h \ge 0$. Along the ray $x_* + r\omega$, the exit length from $P$ is $1/h(\omega)$, with value $+\infty$ when $h(\omega) = 0$.

For a fixed direction $\omega$, set

$$A = h(\omega), \qquad B = h(-\omega), \qquad X_i = v_i \cdot \omega.$$

Then $-B \le X_i \le A$ and $\sum_i \rho_i X_i = 0$. Convexity of $x \mapsto |x|^s$ and comparison with its chord over $[-B, A]$ give

$$\sum_i \rho_i |X_i|^s \le \frac{AB^s + BA^s}{A+B} \le \frac{A^s + B^s}{2}. \tag{S5}$$

The second inequality is equivalent to $(A-B)(A^s - B^s) \ge 0$. If $A + B = 0$, all terms vanish and (S5) holds directly.

Integrating (S5) over the unit sphere and using its symmetry yields

$$\int_{S^{N-1}} h(\omega)^s \, d\omega \geq \kappa_{N,s} \sum_i \rho_i |v_i|^s = \kappa_{N,s} \sum_i \lambda_i a_i^{1-s}, \tag{S6}$$

where the beta integral gives, for any unit vector $e$,

$$\kappa_{N,s} = \int_{S^{N-1}} |e \cdot \omega|^s \, d\omega = \frac{2\pi^{(N-1)/2}\Gamma((s+1)/2)}{\Gamma((N+s)/2)}. \tag{S7}$$

Since $s \geq 1$, the function $a \mapsto a^{1-s}$ is convex on $(0, \infty)$. Jensen's inequality and (S4) therefore imply

$$\sum_i \lambda_i a_i^{1-s} \geq \left( \sum_i \lambda_i a_i \right)^{1-s} = 1.$$

Consequently

$$\int_{S^{N-1}} h(\omega)^s \, d\omega \geq \kappa_{N,s}. \tag{S8}$$

### 3. Radial energy from the maximum point

Let $R(\omega)$ be the exit length from $\Omega$ along $x_* + r\omega$. Convexity ensures that this ray stays in the domain for $0 \leq r < R(\omega)$, and (S3) gives

$$R(\omega) \leq \frac{1}{h(\omega)}. \tag{S9}$$

The test function vanishes before the exit. The fundamental theorem of calculus, followed by weighted Hölder, gives

$$\begin{aligned} M &\leq \int_0^{R(\omega)} |\partial_r f(x_* + r\omega)| \, dr \\ &\leq \left( \int_0^{R(\omega)} |\partial_r f|^p r^{N-1} \, dr \right)^{1/p} \left( \int_0^{R(\omega)} r^{-(N-1)/(p-1)} \, dr \right)^{(p-1)/p}. \end{aligned}$$

The last integral is finite because $p > N$. Evaluating it and rearranging give

$$\int_0^{R(\omega)} |\partial_r f|^p r^{N-1} \, dr \geq \left( \frac{p-N}{p-1} \right)^{p-1} M^p R(\omega)^{N-p}. \tag{S10}$$

Since $N - p = -s$, (S9) implies $R(\omega)^{N-p} \geq h(\omega)^s$, including the case $h(\omega) = 0$. Integrate (S10) in direction, use $|\partial_r f| \leq |\nabla f|$, and apply (S8). This proves

$$\frac{\int_\Omega |\nabla f|^p \, dx}{\|f\|_\infty^p} \geq \left( \frac{p-N}{p-1} \right)^{p-1} \kappa_{N,p-N}.$$

Taking the infimum over all tests and all convex domains of inradius one, and substituting (S7), gives (S2). For a general inradius, scaling restores the factor $R_\Omega^{p-N}$ in (S1).

The two convexity steps in Section 2 require $p - N \geq 1$. Moreover, the radial estimate omits angular derivatives. Equality in an auxiliary geometric inequality therefore would not establish sharpness of (S2) for the original shape optimization problem.

## Problem source and prior asymptotics

The sharp-inradius question is posed in Brasco's Open problems 1, following the superhomogeneous inequality in Proposition 6.3. For a fixed planar domain, the large-$q$ coefficient $8\pi e$ also appears in Bozzola–Brasco, Corollary 3.7 of the published version. A fixed-domain limit alone does not justify interchanging the limit with an infimum over all convex shapes; the uniform Hardy and Adachi–Tanaka argument above provides the needed uniformity.

1. L. Brasco, *On principal frequencies and inradius in convex sets*, Bruno Pini Mathematical Analysis Seminar **9**(1) (2018), 78–101, Proposition 6.3 and (15), p. 99; Open problems 1, p. 100. https://doi.org/10.6092/issn.2240-2829/8945. https://mathematicalanalysis.unibo.it/article/download/8945/8866/28804.

2. F. Bozzola and L. Brasco, *The role of topology and capacity in some bounds for principal frequencies*, The Journal of Geometric Analysis **34** (2024), Article 299. Published Section 3.3, Corollary 3.7, p. 23. https://doi.org/10.1007/s12220-024-01742-2. The author manuscript at https://cvgmt.sns.it/media/doc/paper/6357/bozbra_final.pdf numbers the same result Corollary 3.8, pp. 17–18.